

Evolution of Electronic Check Collection

Two general models of electronic check collection are emerging as a result of the passage of Check 21. Each model has its advantages and disadvantages. In one model, check images including the MICR payment information are transmitted to the paying financial institution. These institutions do not have to rely on multiple image archive providers (with whom they may have no direct contractual relationship) to obtain check images for customer online banking services and back-room operations.

In a second model, only the MICR information is transmitted to the paying financial institution while the check images are stored in remote archives that can be accessed on demand. The MICR information on a check could be transmitted through a dedicated network or possibly the ACH network. A small number of centralized check-image archives could be more cost-effective and might not increase risk appreciably or degrade customer service.

As electronic check collection methods evolve, efficiencies may develop to make one method superior to the other. Notwithstanding, electronic check collection methods will continue to pose certain risks. Frequently-used services that utilize both image and ACH technologies are remotely created checks (RCCs), electronically created payment orders, and remote deposit capture (RDC). Each of these is discussed in the sections that follow.

Remotely Created Checks

A closely related transaction to electronic check conversion, in that there is an authorization to debit an account, is the RCC. ^[13] An RCC does not bear the signature of a person on whose account the check is drawn. In place of the signature, the RCC bears the account holder's printed or typed name or a statement that the account holder authorized the check. ^[14] The account holder can authorize the creation of an RCC by telephone by providing the appropriate information, including the MICR data. Common examples of RCCs are those created by a credit card company, utility company, or telemarketer. RCCs may be processed through the check clearing networks or converted and processed as an ACH debit.

The risk of fraud associated with RCCs is similar to the risk associated with other kinds of debits that post to bank accounts. A fraudster might obtain an account holder's account number by copying that information from one of the account holder's authorized checks, or by tricking the account holder into providing the information over the telephone or the Internet. Once a fraudster obtains the account information, he or she has the data necessary to originate unauthorized RCC transactions through the check collection system or the ACH network. As with all payment systems and mechanisms, a financial institution must also assume responsibility for an effective system of internal controls and ongoing account monitoring related to RCCs.

For RCCs, the check and ACH rules differ as to how an accountholder receives a re-credit for an unauthorized transaction and how the loss is allocated among the participating financial institutions. ACH debits to consumer accounts are governed by applicable ACH rules and by the Electronic Fund Transfer Act and Regulation E. Unauthorized checks posted to consumer accounts are governed by check law, which includes the Uniform Commercial Code (UCC), as enacted in the applicable state, as well as the Expedited Funds Availability Act, as implemented by the Federal Reserve

Board's Regulation CC. In instances when checks are converted to ACH entries, applicable ACH rules apply.

If an unauthorized ACH debit is posted to a consumer's account, Regulation E gives the consumer 60 days after an institution transmits to the consumer a periodic account statement to report that the ACH debit was unauthorized. Regulation E imposes obligations on the consumer's financial institution with respect to error resolution procedures and refunds of unauthorized payments. When a consumer receives a refund for an unauthorized ACH debit, ACH rules permit the consumer's financial institution to recover the amount of the unauthorized payment by returning the debit item to the originating financial institution within the time permitted.

In the case of checks, a financial institution may not charge a customer's account for a check that is not properly payable from that account. The customer has a right to a re-credit for an unauthorized check so long as the customer makes the claim within the time frame permitted by the UCC and the account agreement. Unlike Regulation E, the UCC does not contain specific re-credit procedures that a financial institution must follow. With respect to the allocation of losses for unauthorized checks between financial institutions, the risk of loss falls generally on the paying financial institution, which historically has been in the best position to determine the validity of the drawer's signature. Under the UCC, a paying financial institution becomes accountable for a check unless it returns the check by its midnight deadline.^[15] With the exception of an RCC, if a paying financial institution re-credits a customer's account for an unauthorized check, generally it cannot make a claim against a previous financial institution for an unauthorized drawer's signature after the midnight deadline has passed.

In response to the perceived risk of fraud, legal initiatives have shifted the risk related to unauthorized RCCs from the paying financial institution to the bank of first deposit. This shift is based on the theory that, for unauthorized RCCs, the bank of first deposit is in the best position to know its customer (the creator of the RCC) and to determine the legitimacy of its customer's deposits. A UCC revision that reallocates this risk for RCCs has not yet been widely adopted by the states. Among the states that have enacted amendments to the UCC, the definitions and warranties are not uniform in their scope or requirements. Under the pre-existing provisions of the UCC, the paying financial institution, not its customer, is responsible for unauthorized checks. Providing the paying financial institution with the ability to recover against the financial institution that presented the unauthorized RCC can make it easier for customers to obtain re-credits.

The Federal Reserve Board amended Regulation CC effective July 1, 2006, to reallocate the risk of loss resulting from unauthorized RCCs. Under the amendments, any financial institution that transfers or presents an RCC warrants that the person on whose account the check is drawn authorized the issuance of the check in the amount and to the payee stated on the RCC. The warranty applies only to financial institutions and does not directly create any new rights for checking account customers. Also, any financial institution that received an RCC from another financial institution has up to a year to make a claim against the transferring financial institution for an unauthorized RCC. Similarly, the Board amended Collection of Checks and Other Items by Federal Reserve Banks and Funds Transfers Through Fedwire (Regulation J) in 2006 to clarify that the new warranties apply to RCCs collected through the Reserve Banks. In conjunction with Regulation CC, Regulation J shifted the liability for losses attributable to unauthorized RCCs to the depository financial institution where the check is first cashed or deposited.

Because RCCs are cleared in the same manner as traditional checks, and because nothing unique identifies a check as an RCC unless the signature block on the check is

examined, there is currently no efficient way of measuring the volume or use of RCCs.

Electronically Created Payment Orders

An electronically created payment is a new retail payment practice in which a merchant takes payment instructions for goods and services and places them in an electronic template that creates an electronic file for processing through the check clearing networks. Unlike traditional checks or RCCs, electronically created payment orders do not begin with a paper item. However, they are similar to RCCs in that they are typically initiated with Internet or telephone instructions from the consumer and bear no direct evidence of the customer's authorization. Because these transactions are not originally captured from paper check items, the laws and regulations pertaining to check collection do not apply.

Ordinarily, electronic debits that a consumer uses to acquire goods or services are cleared through the ACH network, which includes a transaction code that clearly indicates the nature and source of the transaction. When a financial institution permits the creation of electronic payment orders, substantial risk-management oversight for unauthorized returns and other unlawful activity is lost because the check-clearing networks do not provide the level of technological and organizational controls of those in the ACH network. This lack of systemized monitoring of the electronically created payment orders increases the susceptibility to fraud by Web-based vendors and telemarketers.

The Federal Reserve Banks handle electronic check images only if they were created from an original paper check. On June 15, 2008, the Federal Reserve Banks revised Federal Reserve Bank Operating Circular 3 (Circular 3) ^[16] to clarify that a depository institution that sends an electronic check file to the Reserve Banks is liable for the legitimacy of the items in that file. Reserve Banks only accept applicable liability and offer certain warranties for Check 21 transactions that begin with an original paper check item. Because electronically created payment orders generally are indistinguishable from electronic images of paper checks, collecting banks, such as the Reserve Banks, may not be able to avoid accepting the electronically created payment orders. However, pursuant to the revised Circular 3, the bank that sends the item to the Reserve Bank ultimately assumes liabilities and provides warranties for its legitimacy.

Remote Deposit Capture

Remote Deposit Capture (RDC), the digital processing of paper checks and monetary instruments at remote locations for deposit and clearing through the check (image) or ACH networks, has expanded rapidly in recent years and is being used at financial institutions and at customer locations. ^[17]

Although remote deposit-taking is not a new activity, RDC should be viewed as a new delivery system and not simply as a new service. Prior to implementing RDC, senior management should identify and assess the legal, compliance, reputation, and operational risks associated with the new system. They should ensure that RDC is compatible with the institution's business strategies and should understand the return on investment and management's ability to manage the risks inherent in RDC. Management should incorporate their assessments of RDC systems, including products and services, into existing risk assessment processes.

With RDC, the depository and collecting financial institutions may choose either to send or accept a substitute check or to engage in electronic check presentment (ECP) where data and images captured from the original checks are used to complete payment transactions. RDC includes deposit capture at the financial institution's teller line and backroom processing, at ATMs, and at customer locations. RDC at customer locations allows the customer to make deposits by scanning items on its own premises and sending either the image of the deposit item for processing through the check clearing networks or merely the deposit data for processing and clearing through the ACH network. RDC also may include the electronic capture of deposit information comprised of cash or other items such as electronic deposits made through a remote safekeeping arrangement at the customer location or through another intermediary.

Financial institutions have a greater degree of control over RDC activities deployed at wholly owned or controlled locations. Based on the RDC configuration used and on the customer's operations, RDC at a customer location increases the financial institution's legal, compliance, and operational risks to varying degrees. Legal and compliance risks could be significant depending on the effectiveness of controls and legal agreements that are in place. The use of RDC by international correspondents' customers is increasing. RDC is effectively replacing correspondent cash letter pouch activity. BSA/ AML controls over RDC pouch activity should also cover RDC and should be commensurate with the increased volumes. Operational risks at the customer location include unauthorized access to technology systems and electronic data images, an inability to maintain system compatibility with financial institution systems, ineffective controls over physical deposit handling and storage procedures, inadequate record retention programs, and exposure to money laundering and fraud.

The Management Booklet of the IT Handbook and the FFIEC Bank Secrecy Act/Anti-Money Laundering (BSA/AML) Examination Manual ^[18] provide additional descriptions of risk management processes.

Check Clearing Houses

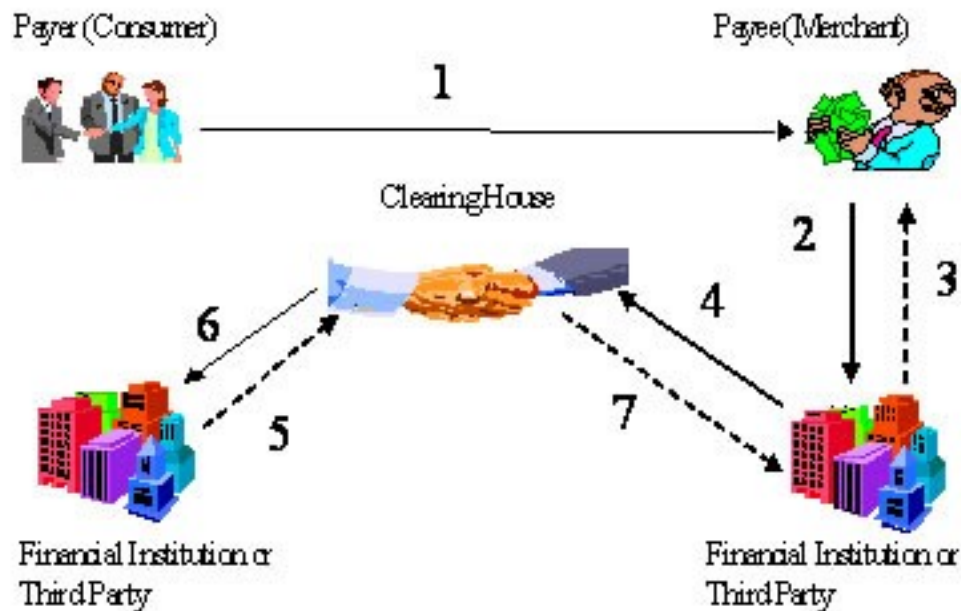
Financial institutions clear and settle checks in different ways depending on whether the checks are "on-us" (checks deposited at the same institution on which they are drawn) or interbank or transit checks (the payer and payee have accounts at different financial institutions). On-us checks do not require interbank clearing or settlement. Interbank or transit checks can clear and settle through direct presentment, a correspondent financial institution, a clearing house, or other intermediaries such as the Reserve Banks.

Under direct presentment, depository financial institutions can present checks directly to the paying financial institution. The paying financial institution may settle with the depository financial institution through a pre-arranged settlement agreement or by sending Fedwire® funds transfers through the Reserve Banks. ^[19]

Correspondent financial institutions, acting on behalf of other depository financial institutions (known as respondents), can settle the checks they collect by using accounts on their books or by using their Reserve Bank reserve account. Smaller depository institutions typically use the check-collection services of correspondent financial institutions or the Reserve Banks.

Financial institutions can also clear checks through a Reserve Bank or through an independent clearing house where they have formed voluntary associations that

establish an exchange for checks drawn on them. With the advent of Check 21, a number of vendors have begun to offer processes and systems for imaging, transferring, archiving, and retrieval of checks. Many financial institutions participating in check clearing houses use the Federal Reserve's National Settlement Service (NSS) to effect settlement for checks exchanged each business day.^[20]



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 2: Check Clearing and Settlement

Figure 2 depicts the typical interbank check clearing and settlement process through a Reserve Bank or clearing house. In step 1 the consumer uses a check to pay a merchant for goods or services. The merchant, after obtaining authorization for the check, accepts the check for payment.^[21] At the end of the day, the merchant accumulates the checks and deposits them with its financial institution for collection (steps 2 and 3). Depending on the location of the paying institution, the funds may not be available immediately. For deposited checks payable at other financial institutions, the merchant's financial institution uses direct presentment for processing or sends the checks to a Reserve Bank, clearing house, or correspondent financial institution (steps 4 and 6). The check or an electronic presentment file is sent to the consumer's financial institution, and the financial institution's account at the correspondent or Reserve Bank is debited (steps 5 and 7).^[22]

Return items are checks that are rejected by the paying financial institution for reasons such as insufficient funds, a closed account, a stop-payment order, fraudulent signature, or failure of the paying financial institution. Return items are a major risk associated with the acceptance of check deposits. The institution that takes a check for deposit may be exposed to credit risk if it releases funds to the depositor and the paying financial institution later returns the check because its customer does not have sufficient funds or

for other reasons.

Regulation CC obligates financial institutions to make deposited funds available for customer withdrawal in accordance with mandatory schedules. Thus, a depository financial institution may be required to make funds available to the customer before an unpaid check is returned to the depository financial institution. When the depository institution receives a return item, it will charge back its depositing customer's account for the item although it had already made the funds available to the customer.

The Automated Clearing House (ACH)

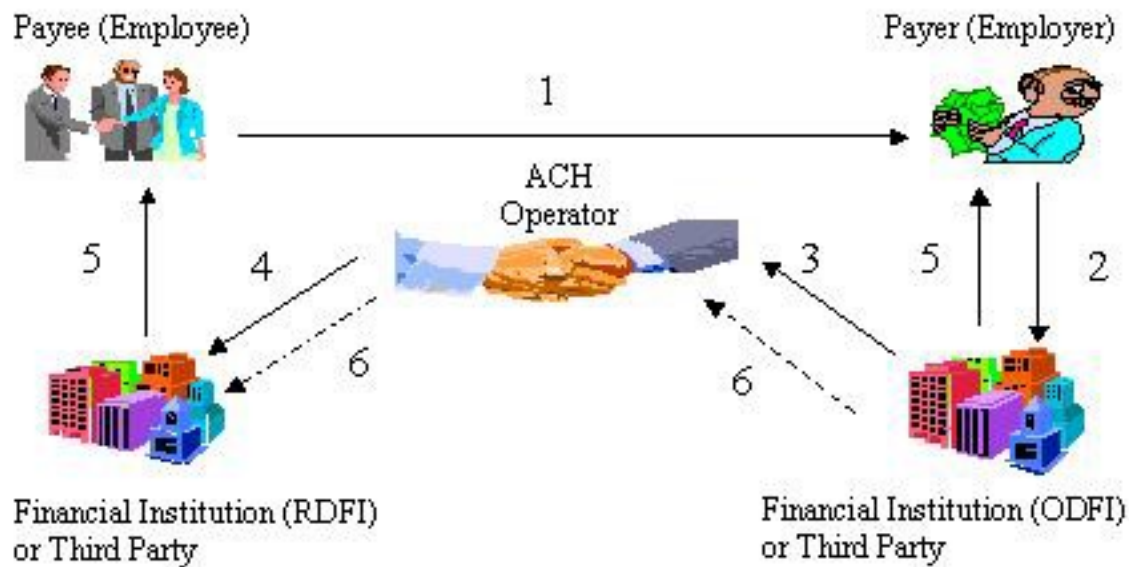
An ACH is an electronic network for the exchange of payment instructions among financial institutions, typically on behalf of customers. ACH transactions are payment instructions to either debit or credit a deposit account. They are batch-processed, value-dated electronic funds transfers between originating and receiving financial institutions. ACH transactions can either be credits, originated by the account holder sending funds (payer), or debits originated by the account holder receiving funds (payee). Financial institutions may contract with third-party service providers to conduct their ACH activities. Unaffiliated independent third parties now generate significant ACH payment activity. NACHA is responsible for the administration, development, and enforcement of the NACHA Operating Rules and sound risk management practices for the ACH Network. ^[23]

The ACH Network

ACH transactions are sent in batches by financial institutions and third-party service providers to ACH operators for processing one or two business days before settlement dates. The ACH operators deliver the transactions to the receiving institutions at defined times. The Electronic Payments Network (EPN), one of the two national ACH operators, is a private processor with a significant share of the national market. ^[24] The Reserve Banks process the remaining share of the market. ACH operators charge a small fee per-transaction to both the originating and receiving depository institutions.

In all ACH transactions, instructions flow from an originating depository financial institution (ODFI) to a receiving depository financial institution (RDFI). An ODFI may request or deliver funds. Transaction instructions and funds are linked using record keeping codes. If the ODFI sends funds, it is a credit transaction. Examples of credit transactions include payroll direct deposit; Social Security payments; dividend and interest payments; and corporate payments to contractors, vendors, or other third parties. If the ODFI requests funds, it is a debit transaction and funds flow in the opposite direction. Examples include collection of insurance premiums, mortgage and loan payments, consumer bill payments, and corporate cash concentration transactions.

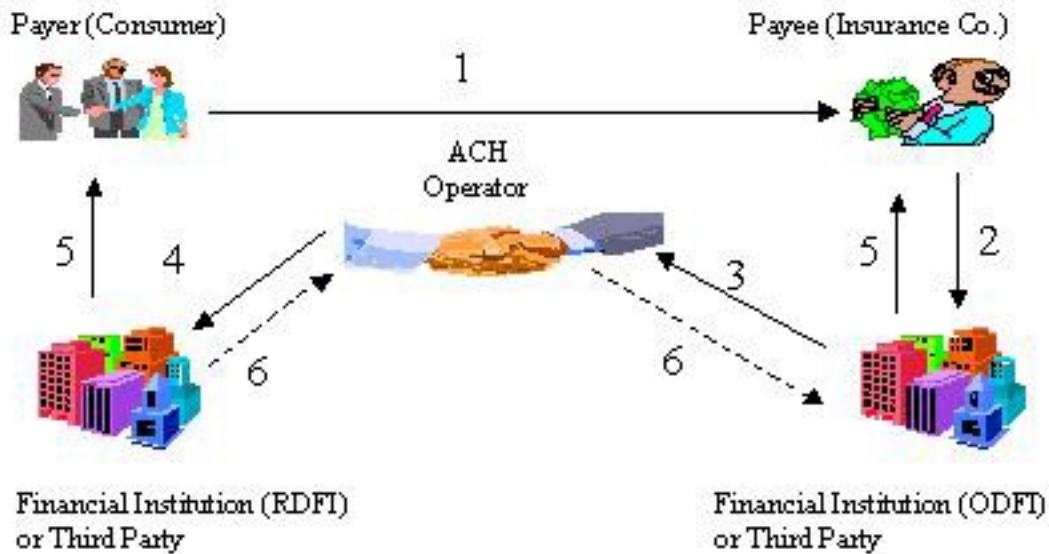
When the ACH files are distributed, financial institutions originating credit payments have a binding commitment for payment to the ACH operator. Settlement for Reserve Bank ACH credit transactions is final at 8:30 a.m. Eastern Time (ET) on the settlement day, when the credits are posted to receiving depository financial institution accounts. Settlement is final for ACH debit transactions, assuming the RDFI has sufficient funds and there are no returns, when posted at 11:00 a.m. ET on the settlement day. ^[25]



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 3: ACH Credit Clearing and Settlement

Figure 3 depicts a typical ACH credit transaction. In this example, the payer is the employer and the payee is the employee. The payee authorizes an employer to deposit his or her paycheck through direct deposit (step 1). The ODFI is the employer's financial institution and the RDFI is the consumer's financial institution. The employer submits its direct deposit payroll ACH files to the ODFI (step 2). The ODFI verifies the files and submits them through the corresponding ACH operator (step 3). The ACH operator routes the transaction to the payee's financial institution, the RDFI (step 4). The RDFI makes the funds available to the payee by crediting his or her account (steps 5). The ACH operator settles the transaction between the participating financial institutions (step 6). If the ACH operator is the EPN, final settlement is made using the Reserve Bank's NSS. If the ACH operator is the Federal Reserve, final settlement is made directly to the financial institution's reserve accounts at a Reserve Bank.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 4: ACH Debit Clearing and Settlement

Figure 4 depicts a typical ACH debit transaction, in this case a recurring monthly insurance premium remittance. The payer sends the ACH payment information and authorization to the payee, in this case an insurance company (step 1). The payee submits this information to its financial institution (step 2), which routes the transaction to an ACH operator (step 3). The ACH operator routes the transaction to the receiving financial institution (step 4). Funds are made available to the payee and the payer's account is debited (step 5). The ACH operator settles the transactions between the participating financial institutions (step 6). Final settlement is performed as described in Figure 3.

An ODFI or an RDFI may outsource ACH processing functions to a third-party service provider, an entity that performs any processing functions on behalf of the ODFI, the originator, or the RDFI, including creation of ACH files or acting as a sending or receiving point. A financial institution may provide the third-party service provider with its Electronic Transaction Identifier (the institution's unique routing number that is used in the ACH network). Third-party senders, customers of the ODFI that provide services to originators, send ACH files on behalf of an originator. ^[26] In a third-party sender model, the ODFI does not have a direct customer relationship with the originator and must rely upon the third-party senders' warranties regarding its originators. The lack of customer knowledge of the originators poses additional risk to the ODFI.

Historically, there was little risk in the ACH system because it was a closed system with recurring transactions and relatively few originators. However, advances in technology

and changes in NACHA Operating Rules resulted in significant changes in the nature and volume of ACH activity, with the most pronounced growth being in nonrecurring payments, potentially increasing the risk of ACH transactions for both financial institutions and their customers. In addition to the primary ACH transactions, retailers and third parties use the now open ACH system for a variety of nonrecurring transactions including:

- ACH check conversion
 - Account receivable (ARC) entries. Many financial institutions operate retail lock boxes for their corporate customers as well as for their own payments collection. Lock boxes receive large volumes of check payments. With ARC, the checks are converted to ACH payments through the transmission of the MICR information on the checks. This data is batch processed for collection through the ACH network. ARC has improved the efficiency of lock-box operations by eliminating the transport of paper checks and increasing the speed of payment collection. While ARC has only been in use since 2001, in 2006 it accounted for 16 per cent of all ACH transactions and was one of the fastest growing segments of the ACH network. Recent statistics, however, indicate that ARC is currently decreasing.
 - Point of Purchase (POP) and Back Office Conversion (BOC) entries. Like ARC entries, POP and BOC entries are created by capturing the check MICR information and sending the transaction through the ACH. The most common application is with checks drawn on consumer accounts. Some retailers and third-party service providers have been converting checks to ACH transactions at the POP or during BOC. BOC was introduced in March 2007 as a new payment solution that allows merchants to collect checks in batches and convert them into debits through the ACH at a central location rather than at the POS. BOC is similar to POP and ARC in that it facilitates the conversion of consumer checks to electronic formats. BOC merely consolidates the electronic conversion process from the individual checkout lines to the back office.
- Internet-originated (WEB) and telephone-initiated (TEL) ACH payments
 - Consumers and retailers can initiate ACH transactions through the telephone and the Internet. These ACH transactions are an alternative to providing a credit card or signature-based debit card number.
- Re-presented check (RCK) entries
 - A physical check that was presented but returned because of insufficient funds may be re-presented as an ACH entry.

NACHA Rule and Product Changes

Over the past few years, NACHA has mandated several important rule changes to expand the use of the ACH network. Some of the more significant changes include:

- Development of a framework to support broader use of international ACH credit and debit transactions and to identify and report international ACH transactions subject to OFAC restrictions. (Effective September 2009^[27]).
- Acceptance of certain business checks for conversion to ACH debits.
- Back-office processing of eligible checks to ACH debits by retailers and billers (BOC entries).
- Use of the ACH network for presentment of bills to consumers.
- Implementation of more stringent network enforcement rules that include more substantial fines for certain violations and permit the ACH Rules Enforcement Panel to direct an ODFI to suspend an originating third party sender.
- Requirement that companies identify themselves within the ACH transaction by the name that is known to, and readily recognized by, the consumer.^[28]

NACHA also requires that every financial institution conduct an annual internal or external audit of compliance with the ACH rules no later than December 31 of each year, and that the audit be made available to NACHA upon request. While the requirements for the "ACH Rule Compliance Audit" do not prescribe a specific methodology, NACHA does identify specific criteria that must be considered during the annual audits (NACHA Operating Rules, Appendix Eight). Financial institutions and third-party service providers should have processes in place to ensure their understanding of, and compliance with, these and future rule and product changes.^[29]

Card-Based Electronic Payments

There is a growing array of card-based electronic payment systems available for retail use. Historically, these payments have been linked to a payee's or payer's existing account relationship with a financial institution. Card-based electronic payments can be defined in three ways, depending on the timing of the payment:

- "Pay Later" payments occur after receiving the goods or services and typically refer to credit payments. A credit card enables a consumer to access a credit line account at a financial institution.
- "Pay Now" payments occur when the goods or services are received and generally are associated with debit payments. Debit card payments are related to an existing transaction account at a financial institution.
- "Pay Before" refers to payments for goods or services with prepaid or stored-value cards, which are loaded with buying power before the purchase of goods or services occurs. The account associated with the pre-paid debit card may be the liability of a financial institution.